PHONOLOGY FORUM 2026 AT KANAZAWA UNIVERSITY

Intersecting families of phonological and extra-linguistic constraints

Evidence from French demonym formation

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ROADMAP

Agenda

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Background

1 Intersecting constraint families

Intersecting constraint families (Zuraw and Hayes, 2017): Phonological outcomes are probabilistically determined by the interaction between two independent dimensions of constraints.

RATES OF VOWEL DEVOICING	constraint applying to high Vs	constraint applying to mid Vs	constraint applying to low Vs
constraint applying to aspirated Cs	/p^{h}i/ → [p^{h}i̥]: x%	/p^{h}e/ → [p^{h}e̥]: y%	/p^{h}a/ → [p^{h}ḁ]: z%
constraint applying to voiceless unaspirated Cs	/pi/ → [pi̥]: w%	/pe/ → [pe̥]: v%	/pa/ → [pḁ]: u%
constraint applying to voiced Cs	/bi/ → [bi̥]: t%	/be/ → [be̥]: s%	/ba/ → [bḁ]: r%

Figure 1: An example of probabilistic vowel devoicing resulting from constraints on vowels (x-axis) and consonants (y-axis) (adapted from Zuraw and Hayes, 2017).

1 French demonyms

A *demonym* is a name for an inhabitant of a place.
The corresponding place name is a *toponym*.

- Formed by suffixation
- Four highly frequent suffixes: *-ois, -ien, -ais, -éen*

(1) French demonym formation (masculine form on the left, feminine form on the right)

a.	Paris [paʁi]	→	Paris**ien** [paʁizjɛ̃]/ Paris**ienne** [paʁizjɛn]
b.	Nice [nis]	→	Niç**ois** [niswa]/ Niç**oise** [niswaz]
c.	Marseille [maʁsɛj]	→	Marseill**ais** [maʁseje]/ Marseill**aise** [maʁsejɛz]
d.	Nancy [nɑ̃si]	→	Nanc**éen** [nɑ̃seɛ̃]/ Nanc**éenne** [nɑ̃seɛn]

1 Factors predicting suffix choice

- **Dissimilative effect** (Plénat, 2008; Roché and Plénat, 2016)
- Interplay between **geography** and phonology (Thuilier et al., 2023)
  - Demonyms of nearby places tend to share the same suffix
- **Psychological reality** of *nasal clash*
  - The specific effect is observed but *dampened* in a "mixed-bag" design

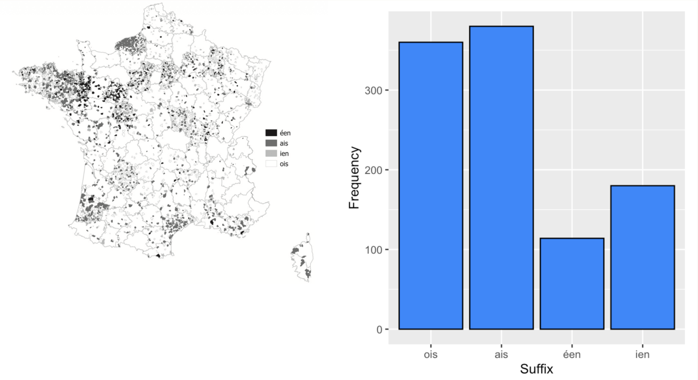

Figure 2: Suffix distribution in France (left) (adapted from Thuilier et al., 2023) and selection rates for stimuli ending in a nasal vowel (right) (adapted from Huygevelde et al., 2025).

1 Current study

Research question

Which cues predict suffix choice in this complex phenomenon?

Aims

- To extend the experiment conducted by Huygevelde et al. (2025) using a broader range of stimuli
- To incorporate a geographical factor into the experiment
  - Tested by comparing two geographically distant participant groups
- To model phonological and extralinguistic variables
- To provide a theoretical account

2

Method

Current experiment: overview

- **Type** Four-alternative forced-choice (4AFC) experiment
- **Platforms** jsPsych (De Leeuw, 2015) / GitHub Pages / DataPipe (De Leeuw, 2023)
- **Data collection** Data were automatically stored in an OSF repository via DataPipe (De Leeuw, 2023)

Two participant groups

Metropolitan French (FF)

- $N = 27$; gender: $F = 22$, $M = 5$; age: $M_{age} = 40.33$ ($SD = 16.48$)
- Recruitment: RISC (UAR 3332 CNRS; `https://www.risc.cnrs.fr/expescience`)

Quebec French (QF)

- $N = 26$; gender: $F = 8$, $M = 18$; age: $M_{age} = 36.88$ ($SD = 10.24$)
- Recruitment: Prolific (`https://www.prolific.com/`: Palan and Schitter, 2018)

Stimuli and carrier sentence I

Stimuli

Jumeu_e [ʒymœ $\top$]

{b, d, f, g, j, k, l, m, n, p, ʁ, s, ʃ, t, v, z, ʒ}

- ***JumeuXe*** was assumed to be phonologically unmarked in French.
- Of the 17 stimuli, 9 were *masculine* and 8 were *feminine*.

Carrier

JumeuXe est une ville (dans la région [REGION NAME]). Paul vient de JumeuXe. Il est __.

- Participants were asked to fill in the blank.
- **Presentation**: Stimuli were presented both orthographically and auditorily; the audio was generated using the Python library `pyttsx3`.

2 Procedure

1. Informed consent
2. Instructions
3. Comprehension test
   - *If all answers were correct:* Proceed to the next task
   - *Otherwise:* Return to the instructions
4. Practice task with existing toponyms (*Paris, Europe*)
5. **Task 1: 4AFC *without any geographical information***
6. **Task 2: 4AFC *with information about the region to which the place belongs***

Figure 3: The four regions labelled in Task 2, each associated with a different linguistic group.

2 Experiment sample

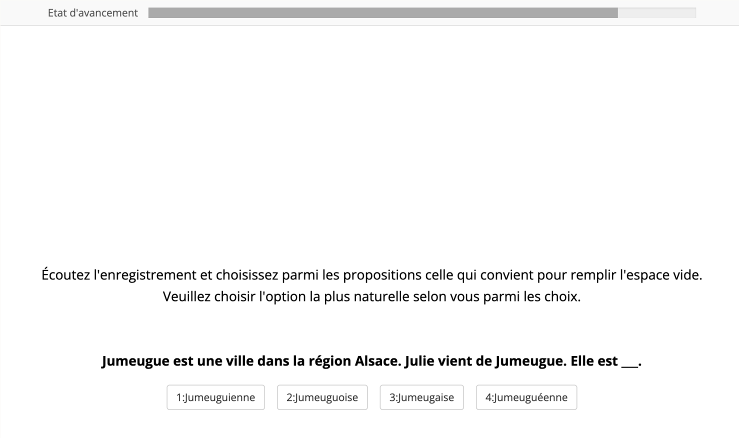

Figure 4: An example from Task 2. The geographical information (“*dans la région Alsace*”) was not presented in Task 1.

Bayesian mixed-effects multinomial logistic regression

- Implemented in R (R Core Team, 2025) using `brms` (Bürkner, 2017)
- Models the probability of *selecting each suffix* as a function of several predictors
  - **Reference category**: *-ois* for FF and *-ien* for QF, because each was the most frequent suffix in its respective group

Model: *RESPONSE* ~ *segment* + *geo_info* + *genre* + *offset(log(rt))* + *(1 | participant)*

- Fixed predictors: **Final segment** (17 levels; reference: *grand mean*), presence of a **geographical label**, and **grammatical gender** (masculine or feminine)
- Offset: Log-transformed reaction time
- Random intercept: Participant

Prior distributions and MCMC settings

- Weakly informative priors (Lemoine, 2019): $b \sim \mathcal{N}(0, 1)$, Intercept $\sim t_3(0, 2.5)$, sd $\sim t_3(0, 2.5)$
- MCMC sampling: (4,000 iterations − 2,000 warm-up iterations) × 4 chains = 8,000 post-warm-up samples
- All $\hat{R}$ values were 1.00, indicating model convergence.

3

Results

Suffix selection: FF

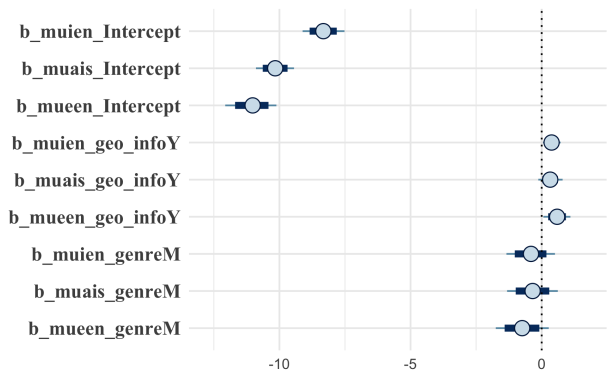

Figure 5: Intervals and densities for the estimated mean selection probability of each suffix relative to *-ois*. The thick navy bars indicate 80% CIs, whereas the thin light-blue bars indicate 95% CIs.

Table 1: Suffix-selection estimates relative to *-ois*[a]

Predictor	β	Error	95% CI	pd
*-ien*_intercept	-9.89	0.36	[-10.59 – -9.20]	≈100%
*-ais*_intercept	-10.90	0.38	[-11.66 – -10.16]	≈100%
*-éen*_intercept	-11.77	0.50	[-12.80 – -10.84]	≈100%

- Suffix-selection tendency (*lexical propensity*): ***-ois* > *-ien* > *-ais* > *-éen***
- Consistent with patterns in attested demonyms reported by Thuilier et al. (2023)

[a] Estimate (β): estimated effect size. 95% credible interval (95% CI): range of plausible parameter values. Probability of direction (pd): probability that the effect has the reported direction.

3 Suffix selection: QF

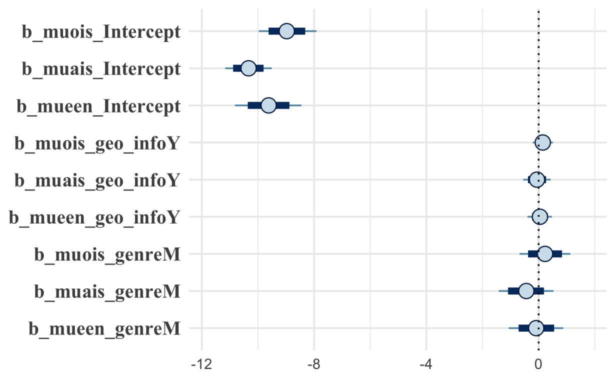

Figure 6: Intervals and densities for the estimated mean selection probability of each suffix relative to *-ien*. The thick navy bars indicate 80% CIs, whereas the thin light-blue bars indicate 95% CIs.

Table 2: Suffix-selection estimates relative to *-ien*

Predictor	β	Error	95% CI	pd
*-ois*_intercept	-8.99	0.53	[-10.00 – -7.93]	≈100%
*-éen*_intercept	-9.66	0.61	[-10.89 – -8.47]	≈100%
*-ais*_intercept	-10.34	0.43	[-11.16 – -9.49]	≈100%

- Lexical propensity: ***-ien* > *-ois* > *-éen* > *-ais***
- Different from the FF result and from patterns in attested demonyms

- In preliminary models, ***grammatical gender did not affect* suffix choice in either variety.** (FF: $\beta = 0.86$, QF: $\beta = -0.32$)

Effect of geographical knowledge

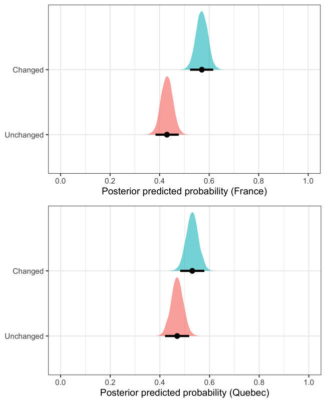

Figure 7: Predicted probability of changing a response between tasks for a given final segment.

Table 3: Both participant groups showed little evidence of systematic response changes.

Condition	95% CI	Post. prob.
FF: Changed	[0.49 – 0.59]	54.3%
QF: Changed	[0.49 – 0.58]	53.4%

- *Presenting the region name* **facilitated** the use of *-ien* and *-éen* among **FF participants**.
  The effect remains uncertain for *-ais*; see Figure 5.
- **No such facilitation was observed among QF participants**.
- FF participants showed a slight tendency to change their responses (Figure 7).

3 Nasal clash effect

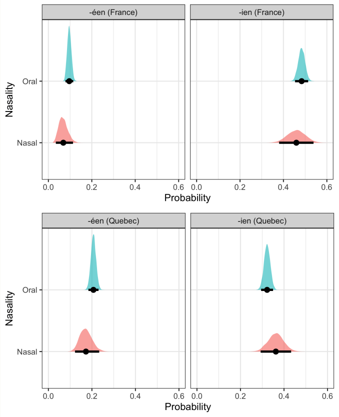

Figure 8: Probability of choosing nasal suffixes in the two varieties.

Table 4: Odds ratios for the OCP effect.

Condition	Odds ratio	95% CI	Post. prob.
FF			
$\frac{\Pr(ien \mid +nasal)}{\Pr(ien \mid -nasal)}$	0.77	[0.56 – 1.01]	97.1%
$\frac{\Pr(een \mid +nasal)}{\Pr(een \mid -nasal)}$	0.73	[0.37 – 1.26]	87.0%
QF			
$\frac{\Pr(ien \mid +nasal)}{\Pr(ien \mid -nasal)}$	1.14	[0.90 – 1.41]	13.1%
$\frac{\Pr(een \mid +nasal)}{\Pr(een \mid -nasal)}$	0.80	[0.56 – 1.11]	90.2%

- A ***modest inhibitory effect*** of nasal clash was observed, *except for QF -ien*.
  Except for QF *-ien*, nasal-final bases multiplied the odds of selecting a nasal suffix by 0.7–0.8.

Interim summary

- ***Strong effect*** of *lexical propensity*
  - FF ***-ois > -ien > -ais > -éen***
  - QF ***-ien > -ois > -éen > -ais***
- ***Noticeable effect*** of *geographical information in FF*
  - Presenting the region name noticeably affected FF participants' responses but not those of QF participants.
  - Possibly due to greater self-monitoring among FF participants?
- ***Modest inhibitory effect*** of *nasal clash*.
  - Nasal suffixes were disfavoured after nasal-final bases, with odds ratios of 0.7–0.8.
  - **Exception: QF *-ien***
    Possibly due to the high lexical propensity of *-ien* in QF.

4

MaxEnt analysis

MaxEnt grammar

Maximum Entropy Harmonic Grammar (MaxEnt: Goldwater and Johnson, 2003; Hayes and Wilson, 2008): A probabilistic approach to constraint-based phonology.

- Constraints are weighted, not ranked

$$\arg\max(\vec{w}) \sum_{i=1}^{K} \ln \Pr(y_i|x_i, \vec{w}) \quad (1)$$

$$H = \sum_{i=1}^{K} w_i C_i \quad (2)$$

$$\Pr = \frac{e^{-H}}{\sum_{i=1}^{N} e^{-H}} \quad (3)$$

Symbols

- w: Constraint weight
- x: Input
- y: Candidate
- H: Harmony
- C: Number of violations

4 Constraints

(2) **FAMILY 1: Phonological**

a. *...[nasal]][...[nasal]...: Assign one violation to a candidate in which the base-final and suffix-initial segments are both associated with [nasal].

(3) **FAMILY 2: Gradually learned lexical propensity** (cf. USE-X: Zuraw and Hayes, 2017)

a. USE-ois: Assign one violation to a candidate that does not contain -*ois*

b. USE-ien: Assign one violation to a candidate that does not contain -*ien*

...

4 MaxEnt tableau for FF

Table 5: Simulation of toponyms ending in a nasal: *loglik = -131.26*
w: weight; violations are indicated numerically; $H = \sum_{i=1}^{K=5} n(``1") \times n(const)$; Pr(obs): observed probability; Pr(exp): predicted probability.

Inp: ...{m, n}e + suf	USE-ois $w = 0.87$	*nasal][nasal $w = 1.09$	USE-ien $w = 1.27$	USE-ais $w = 0.22$	USE-éen $w = 0.00$	H	e^{-H}	Pr(obs)	Pr(exp)
...{m, n}ois			1	1	1	0.13	0.87	49.5%	49.5%
...{m, n}ais	1		1		1	0.78	0.45	25.7%	25.7%
...{m, n}ien	1	1		1	1	0.82	0.44	24.8%	24.8%
...{m, n}éen	1	1	1		1	3.46	0.03	6.5%	6.5%

- Weight optimization and probability calculations were conducted using `maxent.ot` (Mayer et al., 2024)
- The close fit between observed and predicted probabilities indicates that the model reproduces the observed distribution well
- The two constraint families interact
- Despite its lower lexical propensity, *-ais* is favoured over *-ien* because of *nasal][nasal

MaxEnt tableau for QF

Table 6: Simulation of toponyms ending in a nasal: *loglik = -135.96*
w: weight; violations are indicated numerically; $H = \sum_{i=1}^{K=5} n(``1") \times n(const)$; Pr(obs): observed probability; Pr(exp): predicted probability.

Inp: ...{m, n}e + suf	USE-ois $w = 0.96$	*nasal][nasal $w < 0.01$	USE-ien $w = 0.79$	USE-ais $w = 0.00$	USE-éen $w = 0.13$	H	e^{-H}	Pr(obs)	Pr(exp)
...{m, n}ois			1	1	1	0.13	0.87	37.5%	37.5%
...{m, n}ais	1		1		1	0.78	0.45	14.4%	14.4%
...{m, n}ien	1	1		1	1	0.82	0.44	31.7%	31.7%
...{m, n}éen	1	1	1	1		3.46	0.03	16.3%	16.3%

- ***The weight of *nasal][nasal is near zero***
  - Consequently, *-ien* and *-éen* are favoured.

Weight comparison

Table 7: Differences in constraint weights between FF and QF

Variant	USE-ois	USE-ien	USE-ais	USE-éen	*nasal][nasal
FF	0.87	1.27	0.22	0.00	**1.09**
QF	0.96	0.79	0.00	0.13	**<0.01**

- Weights are not directly comparable across fitted models (these models are not intended for typological comparison)
- The largest cross-variety difference is the weight of *nasal][nasal
- Nasal clash creates an intersection that yields different outcome probabilities
  - A similar nasal-clash effect occurs in **Japanese "paired names"** (Kumagai and Kawahara, 2018)
  - Nasal clash also operates weakly across word boundaries, disfavouring adjacent nasals: e.g., *ma**na#ka**na*, **ka**na#ma**na*, *mi**u#mi**ma*, **mi**ma#mi**u*, *mi**ma#hi**na*, **hi**na#mi**ma*

5

Conclusion

5 Conclusion

- Lexical propensity plays a strong role in predicting suffix choice.
  - Native French speakers may use rules of thumb when forming demonyms
- The results confirm an interaction between lexical propensity and probabilistic phonology
- Constraint weights:
  FF: w(USE-ien) $\gg$ w(*...[nasal]][...[nasal]...) $\gg$ w(USE-ois) $\gg$ w(USE-ais) $\gg$ w(USE-éen)
  QF: w(USE-ois) $\gg$ w(USE-ien) $\gg$ w(USE-éen) $\gg$ w(*...[nasal]][...[nasal]...) $\gg$ w(USE-ais)
- Differences in this interaction yield different probabilities of phonological outcomes

5 Future directions: demonym frequency and variation

- Suffix choices for highly frequent demonyms such as *Parisien* and *Européen* are nearly categorical
- Lexical frequency may affect language use (cf. usage-based models: Pierrehumbert, 2001; Todd et al., 2019).
- Although demonym formation is probabilistic, one candidate may dominate when predictability is high

Table 8: Practice-task data for *Paris-SUF*: *loglik = -135.96*

Inp: Paris + suf	USE-ois $w = 0.00$	*nasal][nasal $w = 0.00$	USE-ien $w = 20.25$	USE-ais $w = 0.00$	USE-éen $w = 0.00$	H	e^{-H}	Pr(obs)	Pr(exp)
Parisien	1			1	1	0	1	100%	>99.99%
Parisois			1	1	1	20.25	<0.01	0%	<0.01%
Parisais	1		1		1	20.25	<0.01	0%	<0.01%
Pariséen	1		1	1		20.25	<0.01	0%	<0.01%

6
Backmatter

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- ***Ethics statement***: The experiment was approved by the Aoyama Gakuin University Ethics Review Committee (No. H26-009).
- ***Data availability***: The code and datasets used in this study are available in the Open Science Framework repository (`https://doi.org/10.17605/OSF.IO/RU5KS`).
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Thank you Merci

Questions and comments are welcome in both Japanese and English
Feel free to contact me after the conference.

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